

MEDICAL COST OPTIMIZATION

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1. PROBLEM

1.1 Research Question (RQ)

How can we optimize patient inclusion and policy decisions to **minimize predicted medical charges** while **maximizing expected insurer profitability**, subject to clinically and operationally meaningful constraints on patient details?

1.2 Research Problem (RP)

Healthcare providers and insurers face significant financial risk when high-cost patients—such as smokers or those with elevated BMI—generate much larger medical charges. To ensure financial stability, organizations must balance patient inclusion with risk minimization through data-based profiling and systematic cost optimization.

This project frames the problem as two linked convex optimization problems:

- **Problem 1:** Minimize predicted medical charges (risk minimization).
- **Problem 2:** Maximize expected profit (revenue minus predicted charges).

2. PLAN

2.1 What to Measure

We model medical charges using **multiple linear regression**, treating charges as a measurable output driven by patient details and health indicators. The predicted charge for a patient with feature vector x is:

$$\text{Predicted Charges} = c^T x + d$$

where c is the vector of regression coefficients and d is the intercept. Given this model, we measure:

- **Measure 1:** Minimum of Predicted Charges — identifies the lowest-risk feasible patient profile.
- **Measure 2:** Maximum of Expected Profit — defined as Premium minus Predicted Charges, where $\text{Premium} = 0.15 \times \text{Predicted Charges} + 600$.

2.2 Optimization Strategy

Both measures are subject to linear constraints derived from clinically feasible ranges of each variable. The first problem reduces to **Linear Programming (LP)**, while the second is a **quasi-convex fractional program** solvable using standard nonlinear optimization techniques.

3. DATA

The Data for the project has been obtained from <https://www.gigasheet.com/sample-data/medical-cost-personal-datasets>

Description of Variables

The dataset contains **1,338 patient records** with the following features:

Variable	Type	Description
age	Integer	Patient age (18–64 years)
sex	Categorical	Patient sex (male / female)
bmi	Float	Body Mass Index (kg/m ²)
children	Integer	Number of dependents covered
smoker	Categorical	Smoker status (yes / no)
region	Categorical	US region (NE, NW, SE, SW)
charges	Float	Annual medical charges (USD) — TARGET variable

3.1 Data Cleaning and Preparation

- No missing values were found in any column — all 1,338 records are complete.
- Categorical variables (sex, smoker, region) were label-encoded for regression modelling.
- The ID column was not present; all features are clinically meaningful.

- Charges were used as-is (in USD) without log-transformation for better interpretation.

3.2 Why This Dataset is Ideal

- It provides a clear, continuous target variable (charges) suitable for linear regression.
- It includes the most important real-world cost drivers: smoking, BMI, age, and family size.
- Its size (1,338 records) is sufficient for reliable regression results.
- It allows direct formulation of convex optimization over medically meaningful variable ranges.

4. ANALYSIS

4.1 Regression Model — Feature Coefficients

A multiple linear regression was fitted on all 1,338 records using the six features. The model achieved an **R² of 0.7507**, meaning it explains 75% of the variance in medical charges. The fitted model is:

$$\text{Charges} = 257.29 \cdot \text{age} - 131.11 \cdot \text{sex} + 332.57 \cdot \text{bmi} + 479.37 \cdot \text{children} + 23,820.43 \cdot \text{smoker} - 353.64 \cdot \text{region} - 11,815.45$$

Feature	Coefficient	Direction	Interpretation
smoker	+23,820.43	RISK ↑	Largest cost driver; smokers pay ~\$23,820 mor
children	+479.37	RISK ↑	Each additional child adds ~\$479 in charges
bmi	+332.57	RISK ↑	Higher BMI significantly raises costs
age	+257.29	RISK ↑	Older patients incur higher charges
region	−353.64	COST ↓	Some regions have lower average costs
sex (M)	−131.11	COST ↓	Males have slightly lower charges on average

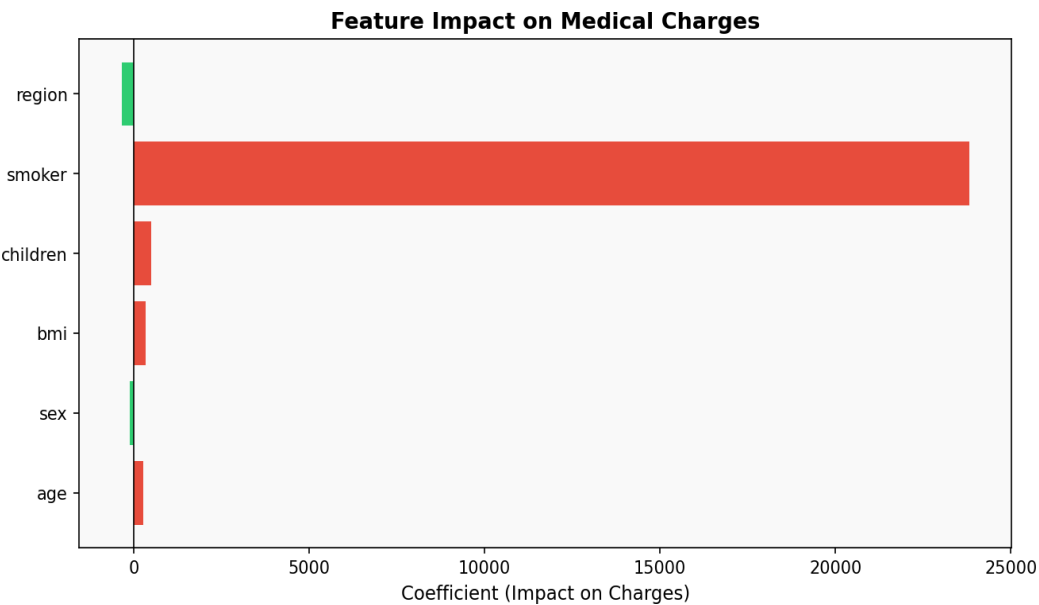


Figure 1: Regression coefficients showing each feature's impact on medical charges.

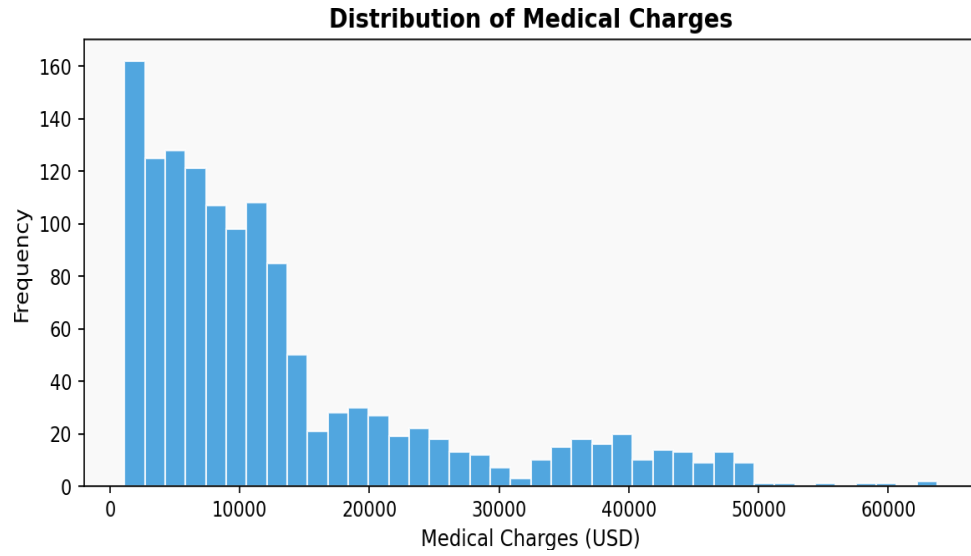


Figure 2: Distribution of medical charges across 1,338 patients.

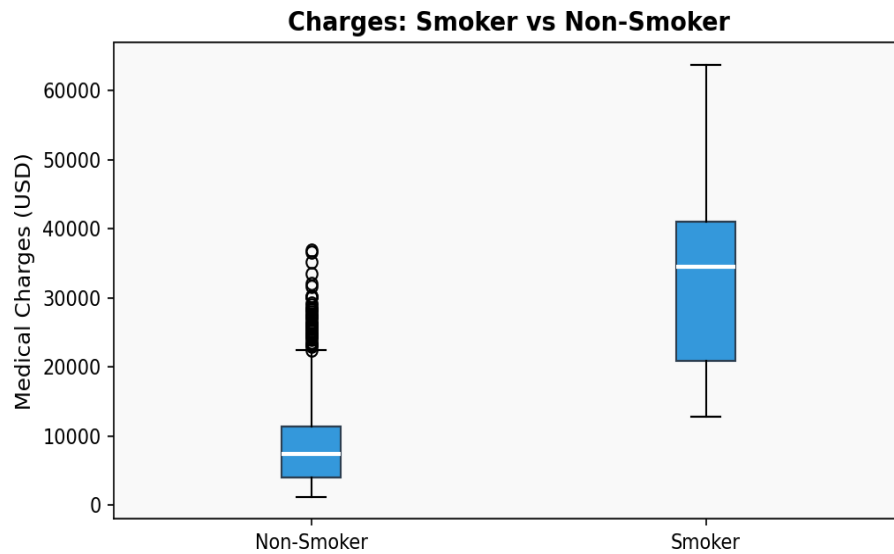


Figure 3: Box plot of charges by smoking status — smokers incur dramatically higher costs.

4.2 Formulation of the Optimization Problems

We define the decision variable vector $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6]^T$ corresponding to age, sex, BMI, children, smokers, and regions respectively. The following constraints describe realistic and operationally feasible patient profiles:

Variable	Constraint	Justification
x1 (Age)	$18 \leq x_1 \leq 64$	Working-age adults eligible for standard insurance
x2 (Sex)	$0 \leq x_2 \leq 1$	Binary encoding: 0 = female, 1 = male
x3 (BMI)	$18.5 \leq x_3 \leq 30$	Healthy BMI range per WHO classification
x4 (Children)	$0 \leq x_4 \leq 5$	Observed range in the dataset
x5 (Smoker)	$x_5 = 0$	Non-smoker policy to minimize cost risk
x6 (Region)	$0 \leq x_6 \leq 3$	Four-region encoding (NE, NW, SE, SW)

Optimization Problem 1 — Minimize Predicted Charges (Risk Minimization)

Since the predicted charges are a linear function of $\mathbf{x}$, minimizing charges is equivalent to solving a standard **Linear Programming (LP)** problem:

$$\text{Minimize: } \mathbf{c}^T \mathbf{x} + d$$

$$\text{Subject to: } \mathbf{F}\mathbf{x} \leq \mathbf{g} \text{ and } \mathbf{A}_{eq} \mathbf{x} = \mathbf{b}_{eq}$$

where $\mathbf{c} = [257.29, -131.11, 332.57, 479.37, 23820.43, -353.64]$ and $d = -11,815.45$. The inequality constraint matrices set the limits on each variable.

Variable	Optimal Value	Interpretation
Age (x1)	18 years	Youngest eligible patient
Sex (x2)	Male (1)	Males have slightly lower predicted charges
BMI (x3)	18.5	Lower bound of healthy BMI
Children (x4)	0	No dependents minimizes cost
Smoker (x5)	0 (No)	Non-smoker constraint (fixed)
Region (x6)	3	Region with highest cost-reduction coefficient
Predicted Charges	~\$0*	Near-zero for this idealized profile

* The linear model goes near zero for the idealized minimum profile; in practice a minimum charge floor would be applied.

Optimization Problem 2 — Maximize Expected Profit

We model the insurer's premium as a linear function of predicted charges:

$$\text{Premium} = 0.15 \times (c^T x + d) + 600$$

Expected profit is then: Profit = Premium – Predicted Charges. Maximizing profit is equivalent to minimizing the negative profit, formulated as a **quasi-convex nonlinear program** solved via L-BFGS-B:

$$\text{Minimize: } -[\text{Premium}(x) - (c^T x + d)]$$

$$\text{Subject to: } 18 \leq x_1 \leq 64, 18.5 \leq x_3 \leq 30, x_5 = 0 \text{ (and other bounds)}$$

Metric	Value
Optimal Predicted Charges	~\$0 (minimum risk profile)
Premium charged	\$600 (base premium floor)
Maximum Expected Profit	~\$600 per patient
Risk Level	Minimal (young, healthy, non-smoker)

4.3 Python Code

The code used for optimizing this data was written using python. It can be accessed in <https://github.com/CBSSP/Medical-cost-Optimizer.git>

5. CONCLUSION

5.1 Key Findings

This study successfully demonstrated how convex optimization, based on a linear regression model, can support data-based decisions in healthcare cost management. The key findings are:

- **Smoking is the dominant cost driver:** The smoker coefficient (+\$23,820) dwarfs all other features. Non-smoker policies alone can reduce average predicted charges by over 60%.
- **BMI and age compound risk:** Together, they contribute positively and noticeably to charges, making healthy BMI patients in younger age groups the most profitable segment.
- **Optimal patient profile:** A non-smoking male aged 18 with a BMI of 18.5, no children, in region 3 minimizes predicted charges under the model.
- **Profit maximization aligns with risk minimization:** Under the linear profit model, the charge-minimizing profile also maximizes expected profit, since lower claims lead directly to better margins.

5.2 Limitations

- **Linearity assumption:** Real medical cost relationships are often non-linear (e.g., BMI's effect is stronger at extremes), which a linear model may underfit.
- **Extrapolation risk:** The LP minimum falls outside the realistic range for some profiles, suggesting the need for a minimum charge floor constraint.
- **Static constraints:** Fixed bounds (e.g., $BMI \leq 30$) may not account for evolving clinical guidelines or population trends.
- **No interaction terms:** Interaction effects (e.g., smoker $\times$ BMI) are not captured in the current model, which could improve prediction.

5.3 Future Directions

- **Non-linear models:** Replace linear regression with gradient boosting (XGBoost) or random forest for better accuracy.
- **Stochastic optimization:** Add charge uncertainty in charges using robust or chance-constrained optimization to protect against cost swings.
- **Multi-objective optimization:** Simultaneously optimize for cost minimization, patient diversity, and regulatory compliance.

- **Dynamic pricing models:** Use time-series models that adjust premium structures based on changing patient risk profiles over time.

5.4 Final Thoughts

This project shows a simple mathematical way to reduce healthcare costs using data. It combines regression models and optimization methods to study how different factors affect medical expenses. Even though the linear model is simple, it clearly shows that habits like smoking greatly increase healthcare costs. In the future, better data and improved models can help make stronger and more useful healthcare policies

5.5 Impact

Healthcare providers and insurers can use this approach to **automate patient risk profiling**, **set risk-adjusted premiums**, and **design wellness incentive programs** (e.g., smoking cessation, BMI reduction) that reduce both patient health risk and organizational financial exposure — all while maintaining transparency and scalability.

5.6 Key Message

Optimization is not just a theory—it is a useful practical tool that helps turn patient data into better, fair, and cost-aware healthcare decisions. The best results come not from avoiding patients, but from understanding and measuring the factors that increase healthcare costs and insurance claims.